Chunlai Long

Multimodal Machine Learning | Numerical Algorithms | HPC

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Experience

Algorithm R&D - Multimodal Learning and Training Engineering *2026.04 - Present*

sEMG hand-pose estimation; FM_PULSE, vEMG2Pose Benchmark, SHM, and Fusion2Pose

- Rebuilt training and real-time inference into a configurable Python/Hydra/CLI platform, supporting force, pose, and gesture multitask learning, uncertainty-weighted losses, reproducible experiments, and modular ONNX inference.
- Diagnosed and corrected a hidden 0.7-1.5 s EMG-Manus timing offset and cross-day clock-domain failures; validated the pipeline across about 850 raw NPZ files, safely isolating one physically corrupted file.
- Established controlled GT A/B, tracking-vs-regression, cross-subject, and cross-device evaluation protocols; achieved 16.54° v1.0 regression test MAE and delivered SHM training, evaluation, checkpoint, and two-stage ONNX workflows.
- Built a 34,163-sample, three-seed training pipeline with real-batch preflight, GPU queuing, checkpoint/resume, and S3 Range GET diagnostics; isolated shared-cache failures instead of misclassifying source objects as corrupt.
- Designed two causal bilateral Vision+sEMG+IMU architectures and processed 368 videos (23.6 h, 2.53M frames) with sequential decoding and FP16 feature caching; Pose-State Query reached 22.62° current test MAE, 2.78° below Chord-MulT, with 90+ contract and integration tests.

Strategy Algorithm Intern, Baidu Mobile Ecosystem Data R&D *2025.09 - 2025.12*

Data Strategy Group

- Applied an AlphaEvolve-style program-evolution framework to domain anomaly detection and phishing-site identification, covering program databases, LLM ensembles, prompt sampling, and evaluator pools.
- Optimized detection models on full-site data and achieved 98% precision in domain anomaly detection.

Selected Research

Multiscale Algorithms for Electromagnetic Models *2020.01 - 2024.01*

- Developed HOMO-FDTD for periodic electromagnetic structures by coupling unit-cell homogenization with time-domain FDTD, reducing grid and compute requirements while preserving physical consistency.

Electromagnetic Wave Propagation in Plasma Sheaths *2024.04 - Present*

- Proved temporal-spatial second-order convergence for three ADE-FDTD schemes under Maxwell-Drude and Maxwell-Lorentz models, deriving error estimates and practical stability criteria.

Education

Academy of Mathematics and Systems Science, Chinese Academy of Sciences *2021.09 - 2026.07*

Ph.D. Candidate in Computational Mathematics | GPA: 3.73

Shihezi University *2017.09 - 2021.06*

B.S. in Mathematics and Applied Mathematics | GPA: 3.68

Technical Skills

- **Machine learning:** TDS/TCN, Transformers, cross-attention, causal masking, multimodal fusion, uncertainty modeling, kinematic FK, multitask learning, Hydra, ONNX.
- **Engineering and HPC:** Python, C/C++, Linux, CUDA, MPI, mmap, S3 data pipelines, checkpoint/resume, multi-seed training, GPU scheduling, unit/contract/integration testing.
- **Numerical computing:** FDTD/ADE-FDTD, finite elements, homogenization, numerical linear algebra, convergence analysis, LAPACK/ScaLAPACK, PETSc, BLACS.

Publications and Awards

- Four manuscripts in preparation on CPDIM/L-DIM convergence for Maxwell-Drude/Maxwell-Lorentz systems and multiscale HOMO-FDTD simulation.
- First Prize, 11th and 12th National College Student Mathematics Competition (Mathematics B); university-level First Prize in Mathematical Modeling; university scholarship; CET-4/CET-6; Teacher Qualification Certificate.